

[Chapter 14](#)—another form of Platonicity. I could not find anyone with depth and scientific technique who looked at the world of randomness and understood its nature, who looked at calculations as an aid, not a principal aim. It took me close to a decade and a half to find that thinker, the man who made many swans gray: Mandelbrot—the great Benoît Mandelbrot.

* The nontechnical (or intuitive) reader can skip this chapter, as it goes into some details about the bell curve. Also, you can skip it if you belong to the category of fortunate people who do not know about the bell curve.

* I have fudged the numbers a bit for simplicity's sake.

* One of the most misunderstood aspects of a Gaussian is its fragility and vulnerability in the estimation of tail events. The odds of a 4 sigma move are twice that of a 4.15 sigma. The odds of a 20 sigma are a trillion times higher than those of a 21 sigma! It means that a small measurement error of the sigma will lead to a massive underestimation of the probability. We can be a trillion times wrong about some events.

† My main point, which I repeat in some form or another throughout Part Three, is as follows. Everything is made easy, conceptually, when you consider that there are two, and only two, possible paradigms: nonscalable (like the Gaussian) and *other* (such as Mandebrotian randomness). The rejection of the application of the nonscalable is sufficient, as we will see later, *to eliminate a certain vision of the world*. This is like negative empiricism: I know a lot by determining what is wrong.

* Note that variables may not be infinitely scalable; there could be a very, very remote upper limit—but we do not know where it is so we treat a given situation as if it were infinitely scalable. Technically, you cannot sell more of one book than there are denizens of the planet—but that upper limit is large enough to be treated as if it didn't exist. Furthermore, who knows, by repackaging the book, you might be able to sell it to a person twice, or get that person to watch the same movie several times.

† As I was revising this draft, in August 2006, I stayed at a hotel in Dedham, Massachusetts, near one of my children's summer camps. There, I was a little intrigued by the abundance of weight-challenged people walking around the lobby and causing problems with elevator backups. It turned out that the annual convention of NAFA, the National Association for Fat Acceptance, was being held there. As most of the members were extremely overweight, I was not able to figure out which delegate was the heaviest: some form of equality prevailed among the very heavy (someone much heavier than the persons I saw would have been dead). I am sure that at the NARA convention, the National Association for Rich Acceptance, one person would dwarf the others, and, even among the superrich, a very small percentage would represent a large section of the total wealth.